

MOUNTAINS OF THE MOON UNIVERSITY
UNIVERSITY EXAMINATION
ACADEMIC YEAR 2021/2022
SCHOOL OF INFORMATICS AND COMPUTING
DEPARTMENT OF COMPUTER SCIENCE
BACHELOR OF COMPUTER SCIENCE (BCS)
YEAR ONE
SEMESTER ONE

PAPER: BCS 1102 Discrete Mathematics

Date: **Wednesday 18/05/2022** Duration: **3 Hours** Time: **09:00 AM – 12:00 AM**

INSTRUCTIONS

- i) *Section A is Compulsory and takes 40 Marks*
- ii) *Answer any three questions in Section B*
- iii) *All questions in Section B carry equal marks*
- iv) *Use SWI-Prolog for any associated practical question*

Section A (40 Marks) (Logic Theory)

- a) Define the following terms **(02 Marks Each)**

- i) predicate
- ii) proposition

- b) Simplify the following Logical expressions with the use of truth table.

- i) $\neg (\neg p \wedge q) \leftrightarrow (\neg q \vee r)$ **(04 Marks)**
- ii) $\neg (p \wedge q) \rightarrow (\neg p \vee \neg q)$ **(04 Marks)**

- c) Look carefully at the sequence of symbols to find the pattern.

- i) Select number for the correct pattern. **(04 Marks)**



- ii) Select number for the correct pattern. **(04 Marks)**



d) (SWI-Prolog Practical Predicate logic)

The file provided (*Family_Relationships.pl*) is concerned with family relationships amongst a group of people. The clauses shown comprise 10 facts defining the *mother/2* predicate, 9 facts defining the *father/2* predicate and 6 clauses defining the *parent/2* predicate. Use it to answer the following questions.

i) Extend the program given by devising rules to define each of the following. Load your extended program and test it.

(a) *child_of(A,B)* (02 Marks)

(b) *grandfather_of(A,B)* (03 Marks)

(c) *grandmother_of(A,B)* (03 Marks)

(d) *great_grandfather_of(A,B)* (02 Marks)

ii) Give the results of running the following goals

(a) *?- child_of(X,ann).* (02 Marks)

(b) *?- grandfather_of(A,caroline).* (02 Marks)

(c) *?- grandmother_of(B,caroline).* (02 Marks)

(d) *?- great_grandfather_of(C,caroline).* (02 Marks)

(e) *?- ancestor(louise,Desc).* (02 Marks)

Section B (60 Marks)

Question One (Boolean Algebra)

a) Simplify the Boolean expressions:

(i) $(X + Y)(X + \bar{Y})(\bar{X} + Z)$ (03 Marks)

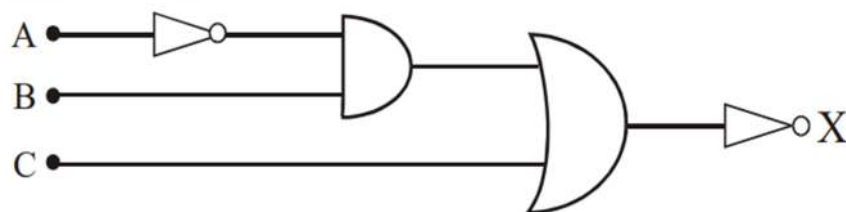
(ii) $XYZ + X\bar{Y}Z + XY\bar{Z}$ (03 Marks)

b) $X = B\bar{C} + B\bar{D} + AB + AD + AC + C\bar{D}$ Minimize the expression by use of the

i) Boolean rules (03 Marks)

ii) Karnaugh map (03 Marks)

C) i) Write the Boolean expression that describes mathematically the behavior of logic circuit shown in figure below. (03 Marks)



ii) Use a truth table to determine what input conditions produce a logic 1 output.

(05 Marks)

Question Two (Set Theory)

- a) Let the universal set be $\mathbf{R}$, the set of all real numbers, and let

$$A = (-1, 0] = \{x \in \mathbf{R} \mid -1 < x \leq 0\} \text{ and } B = [0, 1) = \{x \in \mathbf{R} \mid 0 \leq x < 1\}$$

Find

- i) $A \cup B$ (02 Marks)
 - ii) $A \cap B$ (02 Marks)
 - iii) $B - A$ (02 Marks)
 - iv) A^c (02 Marks)
- b) Let A and B be two finite sets such that $n(A) = 20$, $n(B) = 28$ and $n(A \cup B) = 36$, find $n(A \cap B)$. (06 Marks)
- c) In a town 85% of the people speak Tamil, 40% speak English and 20% speak Hindi. Also 32% speak Tamil and English, 13% speak Tamil and Hindi and 10% speak English and Hindi, find the percentage of people who can speak all the three languages. (06 Marks)

Question Three (Algorithms)

- a) What is an algorithm? (02 Marks)
- b) The following information is to check greatest of two numbers

get a,b value
check if(a>b) print a is greater
Start
else b is greater
Stop

- i) Re-arrange the steps into a meaningful algorithm clearly identifying the step number (03 Marks)
 - ii) Create a flow chart for the above algorithm (03 Marks)
 - iii) Write a pseudocode for the arranged algorithm in i) above (04 Marks)
- c) With the help of a flow chart, create an algorithm that one can use to check greatest of three numbers A, B, C. (08 Marks)

Question Four (Number Theory)

- a) i) Is 1,234,567,890 divisible by 2? (02 Marks)
- ii) Is $121^{13} - 101^4$ divisible by 2? (04 Marks)
- b) Prove that any product of k consecutive positive integers is divisible by k . (06 Marks)
- c) If n is any integer, prove that n^2+n is always divisible by 2, that n^3-n is always divisible by 3, and that n^5-5n^3+4n is always divisible by 5. For a given prime number, p , can you find a polynomial expression like these that is always divisible by p ? (08 Marks)

Question Five (Counting)

- a) The letters S, T, A, R, P are placed in a hat. A letter is removed one at a time from the hat and is not replaced. This process is repeated until all of the letters have been removed. How many different ways can this be done? **(08 Marks)**
- b) If a club consists of 12 members, how many different arrangements of president, vice-president and secretary are possible? **(06 Marks)**
- c) How many different four-digit natural numbers containing only odd digits are there? **(06 Marks)**

~~~~~ *Success* ~~~~~